

Project Proposal Report/Synopsis

for

Predicting Student Dropout Risks using Machine Learning

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I. Title of the Project

Predicting Student Dropout Risks Using Machine Learning

II. Statement of the Problem

Online courses suffer from high dropout rates due to low engagement, poor monitoring, and delayed interventions. Institutions often react only after students have already dropped out, as they lack tools to continuously analyze behavioural data such as attendance, LMS activity, and assessment performance. There is a need for an automated system that can estimate dropout risk early and present it in an interpretable form for academic advisors.

III. Why this particular topic chosen?

With the rapid growth of online and blended learning, ensuring student retention has become a critical challenge for educational institutions. Predictive analytics on student interaction data can support data-driven decisions, helping faculty target timely interventions and improve overall course outcomes. The topic also offers a good platform to apply modern machine learning, data visualization, and full-stack development skills in a socially relevant context.

IV. Objective and Scope

The main objective is to design and implement a web-based system that predicts individual students' dropout risk based on their academic and engagement features and visualizes these risks for stakeholders. Specific objectives include generating or ingesting student datasets, training and evaluating an ML model, providing an interactive dashboard for risk exploration, and enabling alerting (email) for high-risk cases. The scope is limited to predicting risk for a single institution or course dataset, using structured features and supervised learning; long-term institutional integration, adaptive learning, and cross-institutional benchmarking are considered future extensions.

V. Methodology

The project follows a data-driven software development methodology combining CRISP-DM for analytics and iterative prototyping for the frontend. Steps include synthetic/real data collection, preprocessing (handling missing values, encoding, scaling), model training and evaluation using standard metrics (accuracy, precision, recall, F1), followed by

deployment through a Streamlit web interface. The system is refined iteratively based on validation results and usability feedback, ensuring that both model performance and user experience are addressed.

VI. Process Description

The system workflow consists of four main modules: Data Generation, Model Training, Prediction, and Visualization/Notification.

- In Data Generation, synthetic student records are created with configurable parameters and stored for experimentation.
- In Model Training, the data is preprocessed, split into train/test sets, and an ML model is trained and evaluated, with metrics stored in session state.
- In Prediction, users enter individual student details; the model outputs dropout probability, which is rendered as a horizontal gauge with textual interpretation.
- In Visualization/Notification, feature importance and correlation plots explain model behaviour, and high-risk predictions trigger email alert preparation to stakeholders.

VII. Resources and Limitations

The system requires a standard development machine with Python, Streamlit, scikit-learn, NumPy, pandas, Matplotlib/Seaborn, and supporting libraries installed. It assumes access to student data (or uses the built-in synthetic generator) and an email service configuration for notifications. Limitations include reliance on tabular features only, potential bias if training data is not representative, and the fact that the current deployment targets a single-user/demo environment rather than a large-scale production LMS integration.

VIII. Testing Technologies used

The project uses standard software testing and ML evaluation practices. From the ML side, a hold-out test set is used to compute confusion matrix and classification metrics (accuracy, precision, recall, F1) to validate model performance. From the application side, Streamlit pages and forms are manually tested for correct navigation, state handling, and error messages, while boundary conditions for input fields and email alerts are verified with unit-level checks where applicable.

IX. Conclusion

The project delivers an interactive dropout risk prediction system that combines machine learning, data visualization, and notification mechanisms in a single dashboard. It demonstrates how educational institutions can move from reactive to proactive support by continuously monitoring engagement indicators and receiving early warnings for at-risk students. Future work can extend the system with more advanced models, LMS integration, longitudinal tracking, and explainable AI techniques to further improve trust and adoption.